

# The Three-Center Electron-Repulsion Integral Is an Elliptic Bessel Moment: Genus One at the Third Center in Momentum-Space Slater Theory\*

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Two-center electron-repulsion integrals over Slater-type orbitals close in elementary functions plus the exponential integral, the logarithm, and Euler’s  $\gamma$ —a genus-zero transcendence class—which is why the two-center problem has been solved since Roothaan and Ruedenberg. The genuine three-center integral has resisted a closed form for seventy years, and the standard verdict is that it is simply hard: reduce it to an infinite series or a one-dimensional quadrature and be done. We give it a structural reading instead. Evaluating the two-electron three-center integral  $(XY|XZ)$  over  $1s$  orbitals in momentum space—the Fock projection under which Coulomb–Sturmians become hyperspherical harmonics, the setting native to the Averys’ generalized-Sturmian program—the three centers enter as three *phases* rather than three foci, so the coordinate obstruction (prolate spheroidal coordinates admit exactly two foci) dissolves. **[MEASURED]** The resulting momentum integral reproduces the exact integral (agreement  $1.9 \times 10^{-14}$  against a closed-form Gaussian reference;  $4.3 \times 10^{-7}$  against an independent Slater evaluation), and its angular part collapses to a single spherical Bessel kernel. **[MEASURED / classical]** What remains—the radial content—is a two-scale Bessel moment whose period is a *complete elliptic integral* (verified to 31 digits), living on the genus-one curve  $y^2 = (c_1 k^2 + 1)(c_2 k^2 + 1)$ ; it degenerates to the genus-zero two-center case exactly on the diagonal  $c_1 = c_2$ . **[SYMBOLIC + MEASURED]** The one-mass slice is the Laplace transform of the elliptic differential—an elliptic deformation of the Bessel  $K_0$ , reducing to  $K_0(D)$  when one scale decouples ( $\rho \rightarrow 0$ ) and to the complete elliptic integral  $K$  at zero separation—and satisfies an explicit fourth-order Picard–Fuchs equation with polynomial coefficients (residual  $\sim 10^{-24}$ ), whose characteristic exponents are the four branch points of the curve. **[OBSERVATION]** This places a molecular integral inside the modern theory of Bessel moments and two-mass elliptic Feynman integrals; a targeted literature search finds that connection unmade in either field. **[OPEN]** We do not give the finite closed form, and we show the natural route to it is obstructed: the elliptic-dilogarithm (unequal-mass sunrise) mechanism does not engage, because the modulus Picard–Fuchs source is not an elementary subtopology but lies in the integral’s own Gauss–Manin  $D$ -module—the branch-point boundary that would supply the sunrise’s tadpole vanishes identically. **[OBSERVATION]** The object is therefore a genuinely new elliptic transcendent one level *above* the elliptic-dilogarithm tower, a period of a rank-four irregular connection. **[SYMBOLIC + MEASURED]** That connection is *irreducible*—it is the Fourier–Laplace transform of an irreducible connection, and its monodromy admits no invariant coordinate subspace—so no closed form arises through factorization; **[OPEN]** whether the irreducible object admits a transcendental closed form we leave open. The message is not a solution but a relocation: the third center does not make the radial integral messier within the two-center class, it raises the transcendence one genus, past the reach of the sunrise machinery.

## I. THE TWO-CENTER CLASS, AND THE WALL BEYOND IT

The electron-repulsion integral over Slater-type orbitals on two centers is one of the oldest solved problems in molecular quantum mechanics [? ]. In prolate spheroidal coordinates, with the two nuclei at the foci, the Neumann expansion of  $1/r_{12}$  terminates against the orbital angular momenta, and every integral reduces to elementary functions together with the exponential integral  $E_1$ , the logarithm, and Euler’s constant  $\gamma$ . In the transcendence bookkeeping of the Geometric Vacuum program (Paper 18 [? ]), these are *embedding-tier* constants—the cost of the continuum two-center geometry rather than content of the discrete graph—and, decisively, they sit at transcendence *weight one*, and no

dilogarithm appears, even though the ordered radial integral has the simplex shape that generically produces one. In the direct-Coulomb  $(AA|BB)$  class—and there alone—even the prefactor  $\pi$  cancels against the Coulomb  $4\pi$ , leaving a rational combination of  $\{1, e^{-\lambda R}\}$  whose zeros Lindemann–Weierstrass separates; that is what lets *that block* be *decided* rather than merely counted (Paper 58 [? ]). It does not extend to the three genuinely cross classes, which carry  $E_1$ ,  $\ln$  and  $\gamma$  seeds and remain counted—weight one is not the same as decidable.

The genuine three-center integral has no such story. Prolate spheroidal coordinates are built from exactly two foci; a third nucleus has nowhere to sit. Gaussians escape this because two Gaussians on different centers multiply into a single Gaussian on a third point [? ]—Slater orbitals have no product theorem—and this, not any Geometric-Vacuum-specific limitation, is the seventy-year-old reason Gaussian bases displaced Slater orbitals for polyatomic quantum chemistry. The modern

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\* Paper 59 in the Geometric Vacuum series

state of the art for three-center Slater repulsion integrals (Ref. [? ]) reduces them to a single infinite expansion or a one-dimensional numerical quadrature; the one celebrated closed form in correlated three-electron theory, Fromm and Hill’s analytic evaluation [? ], is a *dilogarithm*—a genus-zero object—and belongs to a different integral topology (the one-center, explicitly correlated master integral), which lacks the structure we are about to exhibit.

This paper asks what the three-center integral *is*, transcendently, and answers: it is an elliptic object. The two-center weight-one closure is a genus-zero fact and it stops at two centers. The third center raises the transcendence one genus, to the elliptic (genus-one) world of Bessel moments and two-mass sunrise Feynman integrals. We establish this constructively, validate every step numerically, place the object in the literature, and mark the finite closed form as open.

*Lineage and scope.* The momentum-space route is not new machinery. Molecular integrals over exponential-type orbitals have long been evaluated by Fourier transformation, molecular orbitals in momentum space going back to Shibuya and Wulfman [? ]: the transform of a two-center orbital product is classical [? ], its reduction to four-dimensional (Fock) hyperspherical harmonics goes back to Niukkanen [? ], and the general one- and two-electron multicenter integrals close, via the Fourier-transform method together with Feynman’s identity, to low-dimensional numerical quadrature [? ]. Momentum-space Coulomb–Sturmians are hyperspherical harmonics under Fock’s conformal projection [? ], the framework the Averys built into the generalized-Sturmian method [? ? ? ] and applied to Coulomb–Sturmian repulsion integrals [? ? ]—the closest prior art to the present calculation, which stops, as all prior work does, at expansion and numerics. What is new here is the reading: that the residual radial object is a specific elliptic Bessel moment, and hence that a molecular integral is an instance of the elliptic-Feynman-integral class. We work throughout with  $1s$  orbitals at a common exponent; the generalization to higher  $(n, \ell, m)$  multiplies the integrand by solid harmonics and shifts exponents but does not change the genus, and we return to this in Sec. ??.

## II. THE MOMENTUM-SPACE REDUCTION

Write the chemist integral for two densities  $\rho_1 = \chi_X \chi_Y$  and  $\rho_2 = \chi_X \chi_Z$ , two two-center  $1s$  densities that share the center  $X$  on different axes. With the Fourier representation of the kernel,  $1/|\mathbf{r}_1 - \mathbf{r}_2| = (2\pi^2)^{-1} \int d^3k e^{i\mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2)} / k^2$ , the integral becomes

$$(XY|XZ) = \frac{1}{2\pi^2} \int \frac{d^3k}{k^2} \tilde{\rho}_1(\mathbf{k}) \overline{\tilde{\rho}_2(\mathbf{k})}, \quad \tilde{\rho}(\mathbf{k}) = \int \rho(\mathbf{r}) e^{i\mathbf{k} \cdot \mathbf{r}} d^3r. \quad (1)$$

Each center now appears only through a phase  $e^{i\mathbf{k} \cdot \mathbf{R}}$ : three centers are three phases, not three foci, and there

is nothing to diagonalize. Equation (??) is exact; the only content is the density Fourier transform  $\tilde{\rho}$  and the  $\mathbf{k}$  integral.

We verify Eq. (??) two independent ways on the reference geometry  $X = (0, 0, 0)$ ,  $Y = (0, 0, 2)$ ,  $Z = (1.5, 0, 0.5)$  (bohr),  $\zeta = 1$ , for which a McMurchie–Davidson evaluation over 8-Gaussian  $1s$  fits gives the ground truth  $(XY|XZ) = 0.204941722$ . [**MEASURED**] Using the closed-form Fourier transform of the Gaussian-fitted density (a product of Gaussians is a Gaussian) in Eq. (??) reproduces the McMurchie–Davidson value on the *same* fits to  $1.9 \times 10^{-14}$ : the momentum formula, the Fourier and Coulomb conventions, and the  $\mathbf{k}$ -quadrature are correct. Independently, computing the *true* Slater density transform (no Gaussian fit) by a one-parameter reduction of the two-center Yukawa product reproduces the ground truth to  $4.3 \times 10^{-7}$ . Both are pinned in the regression suite [? ].

*The angular integral closes to a spherical Bessel kernel.* Writing each Slater factor through  $e^{-\zeta r} = -\partial_\zeta(e^{-\zeta r}/r)$  and the Feynman parametrization of the resulting Yukawa product, the three phases combine into a *single* phase  $e^{i\mathbf{k} \cdot \mathbf{W}}$  with  $\mathbf{W}(s, t) = (t-s)\mathbf{X} + s\mathbf{Y} - t\mathbf{Z}$  ( $s, t$  the two Feynman parameters), so the angular integral is elementary,

$$\int d\Omega_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{W}} = 4\pi j_0(k|\mathbf{W}|) = 4\pi \frac{\sin(k|\mathbf{W}|)}{k|\mathbf{W}|}. \quad (2)$$

The whole integral reduces to a two-dimensional Feynman integral over a one-dimensional radial ( $k$ ) integral,

$$(XY|XZ) = \frac{8}{\pi} \partial_{\zeta_a} \partial_{\zeta_b} \partial_{\zeta_c} \partial_{\zeta_d} \int_0^1 ds \int_0^1 dt \int_0^\infty dk j_0(k|\mathbf{W}|) \times \frac{e^{-D_1 \Delta_1}}{\Delta_1} \frac{e^{-D_2 \Delta_2}}{\Delta_2} \Big|_{\zeta=1}, \quad (3)$$

with  $\Delta_1 = \sqrt{c_1 k^2 + 1}$ ,  $\Delta_2 = \sqrt{c_2 k^2 + 1}$ ,  $c_1 = s(1-s)$ ,  $c_2 = t(1-t)$ , and  $D_1 = |X - Y|$ ,  $D_2 = |X - Z|$ . The  $\partial_{\zeta}$ ’s (a bookkeeping device restoring the Slater from the Yukawa) and the  $(s, t)$  integrations are elementary operations; the transcendental content is entirely in the radial factor. Equation (??) reproduces the ground truth to  $1.8 \times 10^{-6}$  [**MEASURED**], the residual being the finite-order angular and Feynman quadrature.

## III. THE TRANSCENDENCE JUMP: GENUS ZERO TO GENUS ONE

The radial factor of Eq. (??) carries two dispersion factors  $\Delta_1, \Delta_2$  with *different* scales  $c_1 \neq c_2$ . A single such factor is benign. Under Fock’s substitution  $k\sqrt{c} = m_3 \sinh \theta$  it linearizes, and [**MEASURED**,  $6 \times 10^{-18}$ ]

$$\int_0^\infty \cos(kb) \frac{e^{-D\sqrt{c k^2 + m^2}}}{\sqrt{c k^2 + m^2}} dk = \frac{1}{\sqrt{c}} K_0\left(\frac{m}{\sqrt{c}} \sqrt{c D^2 + b^2}\right), \quad (4)$$

a Bessel  $K_0$ : the momentum-space Coulomb–Sturmian object, living on a rational (genus-zero) curve. This is why the two-center engine is weight one.

Two different scales cannot be linearized together. Their product is governed by the algebraic curve

$$y^2 = (c_1 k^2 + 1)(c_2 k^2 + 1), \quad (5)$$

a quartic with four distinct branch points  $k = \pm i/\sqrt{c_1}, \pm i/\sqrt{c_2}$  whenever  $c_1 \neq c_2$ : an *elliptic curve*, genus one. When  $c_1 = c_2$  the branch points collide and the quartic becomes a perfect square—rational, genus zero. The cleanest witness is the period. At  $D_1 = D_2 = 0$  the radial integral is exactly a complete elliptic integral [MEASURED, 31 digits; classical],

$$\int_0^\infty \frac{dk}{\sqrt{(c_1 k^2 + 1)(c_2 k^2 + 1)}} = \frac{1}{\sqrt{c_{\max}}} K(m), \quad (6)$$

with modulus  $m = 1 - c_{\min}/c_{\max}$  (nondegenerate,  $0 < m < 1$ ; prefactor verified for both orderings of  $c_1, c_2$  in `test_paper59_euclidean_dictionary.py`), while on the diagonal  $c_1 = c_2 = c$  it collapses to the elementary  $\pi/(2\sqrt{c})$ . Since  $c_1 = s(1-s)$  and  $c_2 = t(1-t)$  range independently over  $[0, \frac{1}{4}]$ , the full integral (??) is an integral over a *family* of elliptic curves, of modulus  $m(s, t) = 1 - c_{\min}/c_{\max}$ , degenerate only on the measure-zero locus  $s = t$  or  $s = 1-t$ . Every term generated by the  $\partial_\zeta$ ’s is a meromorphic function on the same curve (??), so the whole integral is a period of this elliptic family.

The structural statement is that the embedding tier of the transcendence taxonomy (Papers 18, 34 [? ? ]) is *genus-graded*: genus zero ( $\{E_1, \ln, \gamma\}$ ) at two centers, genus one (elliptic) at three. The two two-center negatives—no  $\pi$ , no dilogarithm—are genus-zero properties; the third center does not climb to a higher rung of the polylogarithm weight tower but leaves that tower for the elliptic one. This is the transcendence-level form of “no coordinate system for three foci,” and it also explains, after the fact, why an attempt to identify the three-center value against dilogarithm constants was structurally doomed: the constants are in the wrong class.

#### IV. THE REDUCTION: AN ELLIPTIC BESSEL MOMENT

The radial factor can be named precisely. Applying Eq. (??) to each factor and Parseval’s theorem, the radial core (at fixed Feynman parameters, before the elementary  $j_0$  and  $\partial_\zeta$  wrapping) is a two-mass Bessel moment [MEASURED, 30 digits],

$$M = \frac{2}{\pi\sqrt{c_1 c_2}} \int_0^\infty db K_0(a_1 \sqrt{p_1^2 + b^2}) K_0(a_2 \sqrt{p_2^2 + b^2}), \quad (7)$$

with  $a_i = 1/\sqrt{c_i}$  and  $p_i = D_i \sqrt{c_i}$ . This is a shifted-argument, two-scale member of the Bessel-moment family; it is not one of the single-scale moments

$\int_0^\infty t^k K_0^n(t) dt$  catalogued in closed form by Bailey, Borwein, Broadhurst and Glasser [? ].

A change of variable  $x = \sqrt{c_1 k^2 + 1}$  on the one-mass slice  $N(D) \equiv M(D, 0)$  puts it in the cleanest canonical form: the Laplace transform of the holomorphic elliptic differential,

$$N(D) = \frac{1}{\sqrt{c_1}} \int_1^\infty \frac{e^{-Dx} dx}{\sqrt{(x^2 - 1)(\rho x^2 + 1 - \rho)}}, \quad \rho = \frac{c_2}{c_1}. \quad (8)$$

This is a single explicit one-dimensional integral of an elementary integrand. A low-dimensional reduction is the expected output of the Fourier-transform method—the same machinery reduces multicenter Slater integrals to two- or three-dimensional quadrature [? ] and three-center integrals to a semi-infinite oscillatory Bessel integral [? ]—so what is new is not the reduction itself but the *elliptic reading* of the integrand it produces, which makes the genus-one structure manifest. It is an *elliptic deformation of the Bessel  $K_0$* : as  $\rho \rightarrow 0$  (the one-mass limit, one scale decoupling)  $N \rightarrow K_0(D)$ , and at  $D = 0$  it is the complete elliptic integral  $K$  of Eq. (??).

Because  $1/\sqrt{Q}$ ,  $Q(x) = (x^2 - 1)(\rho x^2 + 1 - \rho)$ , satisfies the first-order relation  $Qf' + \frac{1}{2}Q'f = 0$ , and the boundary at the branch point  $x = 1$  vanishes ( $Q(1)^{1/2} = 0$ ), integration by parts against  $e^{-Dx}$  (using  $\int_1^\infty x^n e^{-Dx} f dx = (-1)^n L^{(n)}(D)$ ,  $L \equiv \sqrt{c_1} N$ ) yields an exact fourth-order Picard–Fuchs equation with polynomial coefficients [SYMBOLIC PROOF; MEASURED residual  $\sim 10^{-24}$ ],

$$D\rho L'''' + 2\rho L''' + D(1-2\rho)L'' + (1-2\rho)L' - D(1-\rho)L = 0. \quad (9)$$

Its characteristic polynomial factors as  $\rho(s^2 - 1)(s^2 + (1 - \rho)/\rho)$ , whose roots  $s = \pm 1$  and  $s = \pm i\sqrt{(1 - \rho)/\rho}$  are exactly the four branch points of the curve (??): the  $K_0/I_0$  sector from the  $(x^2 - 1)$  pair, the oscillatory  $J_0/Y_0$  sector from the other. (A differential-equation method for molecular two-electron integrals is itself established: Pachucki [? ] evaluates the *two-center* integral from a second-order equation in the internuclear distance, closing it in elementary functions; the genus jump appears here as the step from that second-order, genus-zero equation to the fourth-order Picard–Fuchs equation of an elliptic curve.) A subordination identity casts the same object as a two-scale sunrise integral,  $\propto \iint ds_1 ds_2 (s_1 s_2)^{-1/2} e^{-D_1^2/4s_1 - s_1 - D_2^2/4s_2 - s_2} / \sqrt{s_1 c_1 + s_2 c_2}$ , whose second-Symanzik-type  $\sqrt{\text{linear}}$  form is the genus-one signature and separates only when  $c_1 = c_2$ .

One numerical fact pins the object’s class beyond the period.  $N(D)$  is *non-analytic in  $D$  at  $D = 0$* : it carries a  $D \ln D$  term [MEASURED], so it is not a pure elliptic integral ( $K/E$  are analytic in their parameters). The elliptic  $D = 0$  period together with the  $D \ln D$  non-analyticity are the two fingerprints of an *elliptic  $\oplus$  logarithmic* object—the inhomogeneous (“regulator”) term of the Picard–Fuchs solution associated with the elliptic dilogarithm [? ? ? ]—a correspondence whose reach we

test, and bound, in Sec. ??.

[**SYMBOLIC + MEASURED**] *The kinematics.* Equation (??) is the frequency representation of the two-dimensional Euclidean propagator: with  $G_a(\tau, x) = (2\pi)^{-1} K_0(a\sqrt{\tau^2 + x^2})$  the Green's function of  $-\Delta + a^2$  on  $\mathbb{R}^2$ , its left-hand side is exactly  $(2\pi/\sqrt{c}) G_a(p, b)$  with mass  $a = \zeta/\sqrt{c}$ , Euclidean time  $p = D\sqrt{c}$  and space  $b$ , and the fibre (??) is exactly the two-propagator correlator  $M = 4\pi a_1 a_2 \int_{\mathbb{R}} db G_{a_1}(p_1, b) G_{a_2}(p_2, b)$ : two Euclidean packets of unequal mass, evolved from a point source for Euclidean times  $p_1, p_2$  and overlapped on one spatial slice [**MEASURED**,  $10^{-30}$  and  $10^{-31}$  at five parameter points each; `test_paper59_euclidean_dictionary.py`]. Three things become visible. First, the Feynman parameters are Källén–Lehmann spectral variables:  $c = s(1 - s)$  forces  $a = \zeta/\sqrt{c} \geq 2\zeta$ , so each two-centre density is dispersed over its own two-particle cut and the elliptic modulus  $\rho = c_2/c_1 = (a_1/a_2)^2$  is the squared ratio of the two spectral masses—the genus jump is a mass-splitting phenomenon, and the three cusps are equal masses ( $\rho = 1$ ) and one mass at the heavy end of its cut ( $\rho \rightarrow 0, \infty$ ). Second, the exponential twist is  $a_i p_i = \zeta D_i$ , independent of the Feynman parameter: the family is swept at fixed mass $\times$ time, and the physical point is one Compton wavelength of Euclidean-time displacement, the  $u = D$  endpoint of the source-time superposition  $e^{-D\Delta} \Delta^{-n} = \frac{1}{(n-1)!} \int_D^\infty (u - D)^{n-1} e^{-u\Delta} du$  that converts the Yukawa into the Slater. Third,  $j_0(k|W|)$  is a uniform smearing of the shared spatial coordinate over the composite separation, so the third centre enters *only* through space while the bond lengths set the two Euclidean-time offsets [**MEASURED**,  $1.8 \times 10^{-10}$  for the full position-space fibre].

## V. THE ELLIPTIC-DILOGARITHM ROUTE IS OBSTRUCTED

The unequal-mass sunrise is closed by variation of parameters: a second-order Picard–Fuchs operator in the external variable annihilates the two periods, its inhomogeneity is an elementary subtopology (a tadpole), and the particular solution—an iterated integral of the periods against that elementary source—is the elliptic dilogarithm [? ?]. (The full unequal-mass sunrise has since been closed to all orders in  $\varepsilon$  as iterated integrals on  $\overline{\mathcal{M}}_{1,3}$  by Bogner, Müller-Stach and Weinzierl [? ]—a *Fuchsian* object; the point below is that  $N(D)$  is its irregular Laplace dual and so lies outside that technology.) Running the same machinery on  $N(D)$  is the natural attempt. It does not close, and the reason is structural.

The curve  $y^2 = Q(x, \rho)$  of Eq. (??) moves with the modulus  $\rho$ , not with the Laplace variable  $D$ . Accordingly the two periods  $K(\rho), K(1 - \rho)$  are the homogeneous solutions of the second-order modulus operator [**MEASURED: both annihilated to  $10^{-25}$ ;  $L(0, \rho) =$**

$K(1 - \rho)$  to 19 digits]

$$\mathcal{M}_\rho = \rho(1 - \rho) \partial_\rho^2 + (1 - 2\rho) \partial_\rho - \frac{1}{4}, \quad (10)$$

and they are *not* solutions of the fourth-order operator in  $D$ : a constant gives  $\mathcal{O}[1] = -D(1 - \rho) \neq 0$ , and the single Bessel functions of both sectors,  $K_0(D), I_0(D)$  and  $J_0(\omega D), Y_0(\omega D)$  with  $\omega = \sqrt{(1 - \rho)}/\rho$ , are likewise not  $D$ -solutions [**MEASURED**] (they are coupled, which is why Eq. (??) does not factor into elementary Bessel operators). The elliptic-dilogarithm construction lives on the periods, so it must be run in the modulus  $\rho$ .

There it meets the obstruction. Acting with  $\mathcal{M}_\rho$  under the integral sign is an exact total derivative [**SYMBOLIC PROOF; exact residual 0**],

$$\mathcal{M}_\rho[Q^{-1/2}] = \frac{d}{dx} \left[ g(x) Q^{-1/2} \right], \quad g(x) = \frac{-x(x^2 - 1)}{4(\rho x^2 + 1 - \rho)}, \quad g(1) = 0 \quad (11)$$

so for  $D \neq 0$  the modulus source is  $S(D, \rho) \equiv \mathcal{M}_\rho[L] = D \int_1^\infty e^{-Dx} g Q^{-1/2} dx$ . Because  $g$  vanishes at the branch point  $x = 1$ , differentiating in the modulus produces no branch-point boundary term: the analogue of the sunrise's pinched propagator—the elementary tadpole that would be the inhomogeneity—is absent, and the source stays on the elliptic curve. Quantitatively [**MEASURED**],  $S$  closes exactly and finitely only inside the integral's own Gauss–Manin  $D$ -module,

$$S(D, \rho) = \sum_{k=0}^3 P_k(D, \rho) \partial_D^k L, \quad \deg_D P_k = 2, \quad (12)$$

with held-out residual saturating flat under increase of the fit degree (below  $10^{-25}$  in the regression witness,  $5 \times 10^{-44}$  at driver precision; numerically an exact finite identity); whereas the genuine variation-of-parameters shape—period level  $\{L, \partial_D L\}$  plus an elementary Bessel subtopology  $\{K_0, K_1\}(D), \{J_0, Y_0\}(\omega D)$ —only *approximates* it (residual  $2 \times 10^{-14}$  falling geometrically to  $3 \times 10^{-30}$  across fit degrees, never saturating). The inhomogeneity is in-module, not a simpler subtopology, and variation of parameters against the periods is therefore circular.

[**OBSERVATION**]  $N(D)$  is thus one transcendence level above the sunrise—above the *fully solved* sunrise, not merely its leading elliptic dilogarithm: the all-orders unequal-mass sunrise is now closed as iterated integrals on  $\overline{\mathcal{M}}_{1,3}$  [? ], a period of a *Fuchsian* connection, whereas  $N(D)$  is not. It is not an elliptic dilogarithm of elementary or period-level arguments; it is a period of the rank-four irregular connection defined by Eq. (??), for which the sunrise mechanism supplies no reduction to the elliptic-polylogarithm tower [? ]. [**SYMBOLIC**] The irregularity is explicit at the operator level: at  $D = \infty$  the operator of Eq. (??) admits four formal exponential solutions  $e^{\lambda D}$  with  $\lambda \in \{\pm 1, \pm i\sqrt{(1 - \rho)}/\rho\}$ —its leading Newton edge is a full-width slope-one segment, so it is irregular of Poincaré rank one, the four  $\lambda$  being exactly

the branch points (the  $K_0/I_0$  and  $J_0/Y_0$  Bessel sectors), whereas the sunrise/Legendre period operator is regular singular at infinity ( $\lambda = 0$  only). The two operators therefore sit on opposite sides of the Laplace transform, and the Fuchsian elliptic-polylogarithm machinery of [?] cannot reach  $N(D)$ . Whether  $N(D)$  admits a closed form of any kind is a differential-Galois question—whether the fourth-order operator factors over the differential field generated by the periods  $K(\rho)$ ,  $K(1-\rho)$ —settled below, negatively, at the reducibility level (the operator does not factor); what remains open is only whether a *transcendental* closed form exists at the irreducible level.

[MEASURED] The same irregularity is visible in  $N(D)$  itself, as a *resurgent* structure. Expanded about the Bessel point  $\rho = 0$  ( $N \rightarrow K_0(D)$ ), the one-mass slice has the explicit elementary term series

$$N(D) = \frac{1}{\sqrt{c_1}} \sum_{n \geq 0} (-1)^n \frac{((2n-1)!!)^2}{2^n n!} \rho^n \frac{K_n(D)}{D^n}, \quad (13)$$

each coefficient a rational multiple of an *integer-order* Bessel function—the transcendence does not escalate—but the series is *asymptotic*, diverging factorially (optimal truncation at  $n \sim 1-3$ , a few digits, then unbounded). Equivalently the  $D \rightarrow \infty$  expansion  $N(D) \sim e^{-D} \sum_k b_k D^{-(k+1/2)}$  is Gevrey-one, and its Borel transform is the algebraic function  $\psi(\zeta) = [(\zeta+2)(\rho(1+\zeta)^2 + 1-\rho)]^{-1/2}$  on the very curve (??), singular exactly at the three non-dominant branch points  $\zeta \in \{-2, -1 \pm i\sqrt{(1-\rho)/\rho}\}$ —the Stokes actions linking the  $e^{-D}$  sector to the  $e^{+D}$  and  $e^{\pm i\sqrt{(1-\rho)/\rho}D}$  ones. A Borel–Padé resummation confirms these locations (at  $\rho = \frac{1}{5}$ :  $\zeta = -2$  and  $-1 \pm 2i$ , recovered to  $\sim 1\%$ ). So the residual transcendence of  $N(D)$  is not exotic: it is the resurgence of an elementary Bessel series, its Stokes data pinned to the branch points of the elliptic curve.

[MEASURED] The resurgence, moreover, *closes with algebraic Stokes data*. The Borel transform  $\psi(\zeta)$  carries only square-root singularities, so its amplitude at each is algebraic:  $a_* = 1$  at  $\zeta = -2$ , and  $a_*^2 = -\frac{1}{2} \mp \frac{i}{2} \sqrt{\rho/(1-\rho)}$  at  $\zeta = -1 \pm i\sqrt{(1-\rho)/\rho}$  (matched to the closed-form residue to  $10^{-37}$ , and confirmed to be the large-order growth constants by singularity analysis; the dominant Stokes ray switches from the real to the complex pair at  $\rho = \frac{1}{4}$ ). These amplitudes *are* the Stokes constants connecting the four exponential sectors, so the entire trans-series of  $N(D)$  over  $\{e^{\pm D}, e^{\pm i\sqrt{(1-\rho)/\rho}D}\}$  has *algebraic* connection data. The one genuinely elliptic ingredient enters only at the boundary: at  $D = 0$  the open Laplace thimbles close into the cycles of the curve and  $\sqrt{c_1} N(0) = K(1-\rho)$  exactly (14 digits). So the closed form at the irregular level is explicit:  $N(D)$  is the Borel–Laplace transform of the algebraic differential  $\psi$ , with algebraic Stokes data and an elliptic-period  $K(1-\rho)$  normalization—as far as a genuinely irregular, genus-one object reduces. What remains transcendental is the elliptic period at the boundary and the standard Borel–Laplace resummation, not any new constant.

[OBSERVATION] Structurally this is the genus-one analogue of the classical resurgence of the modified Bessel function  $K_0$ —the archetypal rank-one irregular equation, whose large-argument series has an algebraic Borel transform and elementary Stokes data [? ?]. The natural home for the algebraic-Stokes-versus-transcendental-period dichotomy seen here is Sabbah’s theory of Stokes structures for Fourier–Laplace transforms of algebraic connections [?], in which the Stokes data is local at the irregular point (hence algebraic) while transcendental periods enter through the regular points (hence  $K(1-\rho)$  at  $D = 0$ ). The closest worked instance is the rank-two, genus-zero Bessel/Kloosterman connection treated via irregular Hodge theory [?]; and the only resurgence analysis of the sunrise/banana family we found operates in the modular parameter  $q = e^{2\pi i \tau}$  (the  $|q| \rightarrow 1$  limit), not the Laplace-dual  $D$  [?]. The present genus-one, Laplace-dual reading appears unmade; an absence from a targeted search, not a proof.

*A concrete route from the solved dual.* Because  $N(D)$  is the Laplace transform of the sunrise elliptic differential and the unequal-mass sunrise is now known in full closed form [?], one concrete—if not obviously tractable—route to  $N(D)$  is to Laplace-transform that solution directly, an operation that necessarily leaves the elliptic-polylogarithm class for the Bessel-moment/resurgence one, consistent with the irregular character just established. We have not carried this out.

*The rank-four label is not vacuous: a base-field reducibility scan.* That  $N(D)$  is a period of a genuinely rank-four object—rather than one secretly carrying lower-rank content—is itself a reducibility statement, and it can be tested directly. Two structural facts constrain Eq. (??). First, the operator is *formally self-adjoint*,  $L^* = L$  [SYMBOLIC PROOF], so its differential Galois group lies in the symplectic group  $\mathrm{Sp}_4(\mathbb{C})$ —the signature of a polarized period. Second, its indicial exponents at the regular singular point  $D = 0$  are  $\{0, 1, 1, 2\}$  [SYMBOLIC PROOF]; the double exponent at 1 is the local-monodromy source of the  $D \ln D$  nonanalyticity noted above. A search over the base field  $\mathbb{Q}(\rho)(D)$  then returns *no* factor of order one or two—hence, by self-adjointness, none of order three—with polar locus at the true singularities  $\{0, \infty\}$  [SYMBOLIC, positive-control validated]: there is no hyperexponential solution  $e^{\lambda D} Q(D)$  for any  $\lambda \in \{\pm 1, \pm i\omega, 0\}$ , and the only two order-two factorizations the symbol  $(s^2 - 1)(s^2 + \omega^2)$  admits over  $\mathbb{Q}(\rho)$ —the two Bessel sectors—both fail. The two natural elementary structures are excluded by the exponents alone:  $L$  is neither the symmetric product of its Bessel sectors (which would carry exponents  $\{0, 0, 0, 0\}$ ) nor a symmetric cube. This is strong evidence that Eq. (??) is irreducible over  $\mathbb{Q}(\rho)(D)$ , and hence that the rank-four label is earned. The eigenring sharpens it to a single residual target. The endomorphism algebra of the solution space is trivial,  $E(L) = \mathbb{C}$  [computed at  $\rho = 1/5$ ; **robust across pole windows**], so  $L$  is indecomposable;

and the symplectic form pairs the exponential solution  $y_\lambda$  at  $D = \infty$  with  $y_{-\lambda}$ . A non-degenerate invariant subspace would split  $V = W \oplus W^\perp$  and enlarge the eigenring, so every invariant subspace is *isotropic*; the exponential torus  $T = (\mathbb{C}^*)^2$  confines invariant subspaces to spans of the four lines  $\{y_{\pm 1}, y_{\pm i\omega}\}$ , whose two symplectic pairs (the Bessel sectors) are non-degenerate. A one-dimensional isotropic invariant subspace is a hyperexponential solution, already excluded. Hence  $L$  is reducible over  $\mathbb{C}(D)$  *iff* one of the four “mixed” Lagrangian planes  $\text{span}\{y_{\pm 1}, y_{\pm i\omega}\}$  is invariant—equivalently, *iff* the Stokes matrices at  $D = \infty$  [?] fail to couple the real exponential pair  $\{\pm 1\}$  to the imaginary pair  $\{\pm i\omega\}$ . This is decidable, and it is settled twice over—once in principle, once explicitly.

*In principle.* The reduction is an instance of the Fourier–Laplace transform of  $D$ -modules, where irreducibility is automatic. Let  $\mathcal{L}$  be the rank-one meromorphic connection on  $\mathbb{P}^1$  whose flat sections are  $Q^{-1/2}$ : it is regular singular, with local monodromy  $-1$  at each of the four branch points  $x \in \{\pm 1, \pm i\omega\}$  and trivial monodromy at  $\infty$  (there  $Q^{-1/2} \sim x^{-2}$ ). Having rank one,  $\mathcal{L}$  is irreducible, so its minimal (intermediate) extension  $j_{1*}\mathcal{L}$  to  $\mathbb{A}^1$  is a *simple* holonomic  $D$ -module. (Because every finite local monodromy is  $-1 \neq 1$  and hence has no invariant,  $j_1\mathcal{L} = j_{1*}\mathcal{L} = j_*\mathcal{L}$ , so “the  $D$ -module generated by  $Q^{-1/2}$ ” is unambiguous.) The Laplace kernel  $e^{-Dx}$  makes the radial integral the Fourier–Laplace transform  $\text{FL}(j_{1*}\mathcal{L})$ , and  $\text{FL}$  is an exact auto-equivalence of the category of holonomic  $D$ -modules on the affine line (with its dual coordinate) [?]; an equivalence sends simple objects to simple objects, so  $\text{FL}(j_{1*}\mathcal{L})$  is simple. Its generic rank is the total drop of  $\mathcal{L}$ —one unit at each branch point, where the monodromy  $-1$  leaves no invariant, and none at the regular point  $\infty$ , hence *four*, matching the order of Eq. (??), of which  $N(D)$  is the defining (cyclic) solution. A simple holonomic  $D$ -module of generic rank four restricts to an irreducible connection over the function field, so **the operator of Eq. (??) is irreducible** over  $\mathbb{C}(D)$ , hence over  $\mathbb{Q}(\rho)(D)$ . Simplicity is a property of the connection at the generic point, so it forbids a factor of *any* order—apparent singularities and all; the monodromy computation next verifies it by hand.

*Explicitly.* The same conclusion is exhibited, and the mechanism made visible, by the monodromy. The four exponentials are the branch points  $x \in \{\pm 1, \pm i\omega\}$  of  $y^2 = Q$  in the Borel (Laplace-dual) plane, and the monodromy of  $L$  around  $D = 0$ , computed in the exponential-line basis of Lefschetz-thimble solutions  $\int e^{-Dx} dx / \sqrt{Q}$ , is **[MEASURED; exact-integer certificate]** the integer unipotent matrix

$$M_0 = \begin{pmatrix} -1 & 2 & 2 & 2 \\ -2 & 3 & 2 & 2 \\ -2 & 2 & 3 & 2 \\ 2 & -2 & -2 & -1 \end{pmatrix},$$

the *same* for  $\rho = \frac{1}{5}$  and  $\rho = \frac{1}{3}$  (a topological invariant of the branch-point configuration), with a single  $2 \times 2$  Jor-

dan block—the lone  $D \ln D \log$ —and *no* invariant coordinate subspace. The Stokes matrices therefore *do* couple the real and imaginary exponential pairs, no mixed plane survives, and the reducibility criterion above returns *irreducible*—confirming the structural argument concretely. **[SYMBOLIC + MEASURED]** Consequently  $N(D)$  is a period of an *irreducible* rank-four connection: it admits no closed form through any factorization of  $L$ , and the sole remaining route to a closed form is a genuinely transcendental one at the full rank-four level—the elliptic-polylogarithm /  $\Gamma(2)$  multiple-modular-value structure of Sec. ???. The differential-Galois *reducibility* question is thereby settled (negatively:  $L$  is irreducible); the transcendental closed form is the one problem that remains open [?].

## VI. PLACEMENT IN THE LITERATURE

The object of Eqs. (??)–(??) sits inside a fully developed body of mathematics, applied so far only to problems far from chemistry. Two-mass elliptic Feynman integrals—the sunrise/banana family—are evaluated in terms of elliptic dilogarithms as inhomogeneous solutions of exactly this kind of Picard–Fuchs equation [? ? ?]; their periods are the complete elliptic integrals of Eq. (??), and at special (complex-multiplication) scale ratios their values reach critical  $L$ -values of modular forms [? ? ? ?]. The closest published analog to Eq. (??) is a doubled-argument Bessel moment that Broadhurst evaluated as a product of complete elliptic integrals via a lattice-Green-function contour [?]: the same “two-scale moment  $\rightarrow$  product of  $K$ ’s” mechanism we expect at rational  $\rho$ .

On the chemistry side the treatment is uniformly numerical: three-center Slater repulsion integrals are reduced to infinite series or one-dimensional quadrature [? ?], and the Averys’ momentum-space Coulomb–Sturmian route [?], closest in spirit to ours, likewise stops at expansion and numerics. We could find no paper drawing the bridge in either direction: no chemistry reference names an electron-repulsion integral as a Bessel moment or an elliptic Feynman integral, and no amplitudes reference applies the elliptic machinery to a molecular integral. **[OBSERVATION]** The identification—that the three-center Slater integral, via the Fock projection, is a two-scale Bessel moment on the sunrise elliptic curve—appears to be unmade. This is an absence from a targeted search, not a proof of absence; the residual risk is an obscure single source. **[OBSERVATION]** A second placement datum is arithmetic: the conductor-4 content this family carries (the  $\mathbb{Z}[i]$  theta series  $\theta_3^2$  of Sec. ??;  $G = \beta(2) = L(2, \chi_{-4})$  in the corrected ring) is absent from the established Bessel-moment corpus, which sits at levels/conductors 3, 6, 8, 15 [? ? ?]—a targeted search finds no Catalan-constant appearance in any of these—the nearest  $\chi_{-4}$  appearance being the topological-string sector of the Broadhurst–Dorigoni programme [?

]. The family thus occupies an apparently unclaimed arithmetic slot (conductor 4) in that literature, of a piece with the unmade bridge above. (Reconciliation with the value-level searches of Sec. 4: the conductor-4 arithmetic is a property of the *family* and of the  $D \rightarrow 0$  shadow; the physical integrated value itself shows no measured conductor-4 content at reachable heights.)

## VII. THE MODULAR STRUCTURE AND ITS COSMIC-GALOIS PERIODS

The genus-one curve of Eq. (??) is not an arbitrary elliptic curve: the family over the scale-ratio base is a *modular* family, and its periods are periods of a specific modular curve. Forming the modular parameter from the two periods of Sec. ??,  $\tau(\rho) = iK(\rho)/K(1-\rho)$ , the theta-series modular lambda satisfies [MEASURED,  $10^{-41}$ ; classical]

$$\lambda(\tau(\rho)) = 1 - \rho, \quad (14)$$

an exact, rational—indeed linear—modulus map. The family is therefore the Legendre family, the universal elliptic curve over the modular curve  $X(2) = \Gamma(2)\backslash\mathfrak{H}^*$ , pulled back along  $\lambda = 1 - \rho$ ; it is non-isotrivial ( $j(\rho)$  varies) with three cusps at  $\rho \in \{0, 1, \infty\}$  matching the three cusps of  $\Gamma(2)$  (the cusp  $\rho = 1$  is the coincident-scale diagonal). The periods  $K(1-\rho), K(\rho)$  are the  $\Gamma(2)$  modular periods.

*The modulus is a four-point cross-ratio.* Identity (??) has an elementary source. Under  $u = x^2$  the quartic model reduces  $2 : 1$  to the elliptic curve  $v^2 = u(u-1)(\rho u + 1 - \rho)$ , whose periods are  $K(\rho), K(1-\rho)$  and whose four branch points are  $\{0, 1, \infty, (\rho-1)/\rho\}$ . Their six-element cross-ratio orbit is  $\{(\rho-1)/\rho, 1/\rho, \rho/(\rho-1), \rho, 1-\rho, -1/(\rho-1)\}$  [SYMBOLIC], and it contains *both* period arguments  $\rho$  and  $1-\rho$ ; the Legendre value  $1-\rho$  is one of its members. The elliptic modulus of the three-center integral is therefore the classical modulus of four marked points on  $\mathbb{P}^1$ —the tame four-subspace  $(\tilde{D}_4)$  modulus, equivalently the cross-ratio of the four regular singular points of a rank-two Fuchsian system (the isomonodromic setting whose deformation is Painlevé VI). One caveat fixes the reading: the four points are the branch data of the *two* spectral masses  $c_1, c_2$  through  $\rho = c_2/c_1$  (each mass contributes a  $\pm$  pair, one pair sent to  $\{0, \infty\}$  and the other to  $\{1, (\rho-1)/\rho\}$  by the reduction), *not* the four positions of four scattering centers—the modulus counts the two densities, not the number of nuclei.

Since  $\rho = c_2/c_1$  ranges over all of  $(0, \infty)$  as the Feynman parameters range over  $(0, 1)$ , the parameter integral sweeps the entire real locus of  $X(2)$  and passes through every complex-multiplication fibre. There the periods are  $\Gamma$ -values by Chowla–Selberg [MEASURED,  $\sim 10^{-51}$ ; classical]: at  $\tau = i$  (the fibre  $\rho = \frac{1}{2}$ , *inside* the physical domain),  $K(\frac{1}{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$ ; at  $\tau = i\sqrt{2}$  (discriminant  $-8$ ),  $K = (1 + \sqrt{2})^{1/2}\Gamma(1/8)\Gamma(3/8)/(2^{13/4}\sqrt{\pi})$ . A

molecular three-center integral thus has, at the fibres its own parameter integral visits, periods that are classical  $\Gamma$ -values—across distinct fundamental discriminants, so the arithmetic tracks the fibre rather than a single coincidence.

*The integrated integral is an elliptic Bessel moment.* The full parameter integral pulls back, via Eq. (??), to an iterated integral over  $X(2)$  of period-weighted forms. The fibre, however, carries an irregular exponential (Bessel) twist—it degenerates to the Bessel value  $K_0(1)$  as  $\rho \rightarrow 0$ , which is not a modular period—so the object is properly a *Bessel moment of the Legendre family*, an irregular/exponential period in the sense of Fresán–Sabbah–Yu [? ], of which the  $\Gamma(2)$  multiple modular value in the sense of Brown [? ] (an elliptic polylogarithm [? ? ]) is the regular  $D \rightarrow 0$  shadow. A dedicated evaluator—a fixed Gauss–Legendre tensor grid with a  $\sin^2$  substitution on the smooth bulk and, at the  $(s, t) \rightarrow (0, 0)$  corner, a Duffy angular split followed by the radial substitution  $\rho = \sigma^2$  ( $\rho = s + t$ ) that renders the  $\rho^{3/2}$  corner integrand *analytic* in  $\sigma$  and so restores spectral Gauss–Legendre convergence—gives the collinear value [MEASURED, 18 digits confirmed]  $0.3953557659017139641\dots$  (correct to 18 digits; the 19th is superseded by the certified value below). The earlier  $\sim 16$ -digit figure was not an intrinsic wall but an under-resolution of the corner and of the one wide sub-domain flagged above (both closed here; the corner by  $\rho = \sigma^2$ , the sub-domain by cross-validated Gauss–Legendre and  $\tanh$ – $\sinh$  quadrature); the  $\sim 20$ -digit ceiling was a fixed-fibre-grid artifact, not intrinsic: on the decay-scaled map  $k = Lu/(1-u)$ ,  $L = 1/(\sqrt{c_1} + \sqrt{c_2})$ , a fixed Gauss–Legendre fibre is *spectral* to 40-plus digits (the fibre is not the bottleneck), and a rebuilt spectral evaluator ( $\rho = \sigma^2$  radial +  $\sin^2$  angular corner, kink-free trapezoid split) independently reproduces the value to 16 digits (the two structurally distinct evaluators agree to 16 digits; a naive single-axis outer-grid push produces non-monotonic near-coincidences past that and is not trusted); the residual precision limit is the outer-integral Gauss–Legendre rate ( $\sim e^{-0.55N}$ , complex-singularity-limited), so within the  $(s, t)$  frame a definitive evaluation required a different representation—supplied below by the exact  $(k, w)$  refactorization, Eq. (??), not by the modular route. [MEASURED] A guarded integer-relation search with a same-magnitude decoy control finds the value is *not* a low-height combination of the small disc-4 complex-multiplication ring  $\{1, \pi, K(\frac{1}{2}) = \Gamma(\frac{1}{4})^2/4\sqrt{\pi}\}$  at weight  $\leq 2$ : the relation height is far above low and is matched by the decoy, the signature of no low-height relation—and this leg is decisive, since a weight- $\leq 2$  disc-4 basis is small enough to resolve at the reachable precision. The wider searches, through weight three and against the disc-8-inclusive ring  $\{\pi, K(\frac{1}{2}), P_8\}$ , are only *consistent* with the negative at present: those bases are over-determined at  $\sim 19$  digits—the spurious low-height hits differ from one precision to the next, and are matched by the decoy—a limitation removed below. All of this

is consistent with a genuine (non-classical, multi-fibre) elliptic Bessel-moment period.

[**SYMBOLIC + MEASURED**] Both the precision wall and the pending weight-three decision are closed by an exact refactorization. Writing  $j_0(kb) = \int_0^1 \cos(kbw) dw$  decouples the  $b = s + t$  phase, and the reflection symmetry  $P(s, k) = P(1 - s, k)$  of the collinear packet makes the inner integral real, so the  $(s, t)$  double integral collapses to the *square of a one-dimensional integral*:

$$T2 = \frac{8}{\pi} \int_0^\infty dk \int_0^1 dw \cos(kw) R(k, w)^2, \quad R(k, w) = \int_0^1 \cos(kws) ds \quad (15)$$

In this frame the complex off-axis  $(s, t)$  singularity that produced the  $\sim 13$ – $14$ -digit outer wall does not exist (the  $k$ -integrand is entire in  $w$ ; the  $s$ -integrand is analytic on  $[0, 1]$ ), no fibre is ever evaluated, and the entire precision question is the closed-form large- $k$  tail (a Watson expansion; the Gevrey structure relocates to a truncation floor  $e^{-K/2}$  with the cut  $K$  freely chosen). Six parameter-disjoint runs—two independent parallel configurations agreeing to  $1.7 \times 10^{-67}$ , the identity itself verified against a fully independent two-dimensional evaluation to 69–96 digits—certify

$$T2 = 0.3953557659017139643252292968048475642605639778670821089352542654691$$

(66 digits; a *decomposed* certification—identity, quadrature and tail legs each closed independently, at higher precision than any single end-to-end pipeline; `tests/test_paper59_t2_value.py` pins the identity, the  $K^{-7}$  tail law, and the digit-19 correction of the frozen anchor above, and the certified entry with its full evidence accounting is in the reference artifact, `benchmarks/certified_reference/`). At this precision the guarded, decoy-calibrated integer-relation searches become decisive across the board: the corrected weight- $\leq 3$  ring  $\{\pi, K(\frac{1}{2})^{\pm 1}, G\}$  is **decisively negative** at height  $\leq 10$  (64 working digits, two precisions), as are disc-4 through weight three and disc-8 through weight two at heights up to  $10^4$ – $10^{12}$ , and eight targeted Catalan probes—*no measured conductor-4* ( $\beta(2)/\text{Catalan}$ ) *content exists in the integrated observable at clean heights*. This sharpens the shadow reading (the finite weight-three-requiring- $G$  element belongs to the  $D \rightarrow 0$  shadow, not to  $T2$ ) and answers the Paper 56 seam test T-2 in the negative at the clean-height bound. Two calibration riders: a decision on the dimension-twenty ring was never available at the previously projected  $\sim 32$ – $40$  digits (height budgets scale as  $10^{D/n}$  in the ring dimension  $n$ ), and the negative excludes *clean* closed forms, not large-height ones. The resurgent trans-series program below is untouched—digits and closed form are separate questions; the closed form remains [OPEN].

[**MEASURED**] Two refinements sharpen this. *First*, the fibre precision wall is removable. The edge oscillation of  $j_0(k(s + t))$ —which runs over a  $k$ -range  $\sim 1/\sqrt{c_{\min}}$  as a Fock scale collapses and limited every

fixed- and decay-grid fibre quadrature to  $\sim 16$ – $22$  digits—is eliminated by an *oscillation-free* evaluator: split the radial integral as  $\int_0^K + \int_K^\infty$  and evaluate the oscillatory tail  $\int_K^\infty \sin(kb) e^{-Ak} k^{-n} dk = \text{Im}[z^{n-1} \Gamma(1-n, zK)]$  in closed form ( $z = A - ib$ ,  $A = \sqrt{c_1} + \sqrt{c_2}$ ,  $b = s + t$ ), the tail amplitudes taken from the  $1/k$ -series of the cancellation-stabilised combination  $g(k) = P_1(k)P_2(k) e^{Ak} k^6$  (a single  $\Gamma(-6, zK)$  plus downward recurrence). This reproduces the decay-grid fibre to 40–45 digits at every  $(s, t)$ , the corner included, and reproduces the collinear value to its confirmed digits. With *the fibre no longer the bottleneck*, the residual precision wall of that frame was the *outer*  $(s, t)$ -integral's corner-limited convergence—which held the value near a  $\sim 19$ -digit ceiling there until the  $(k, w)$  refactorization of Eq. (??) dissolved the question—not the fibre. This outer wall is now characterized: of the four corners only  $(s, t) \rightarrow (0, 0)$  is fractionally singular ( $\rho^{3/2}$ , absorbed by the  $\rho = \sigma^2$  Duffy)—the three oscillatory corners ( $b = s + t \neq 0$ ) are integer-leading (leading order  $\rho^2$ ) and carry no fractional-power subtraction [**MEASURED; test\_paper59\_resurgence\_corner.py**]*—so the cap is a weak complex off-axis singularity of  $J$  (its signature a non-monotone Gauss–Legendre rate), not a missing corner map; a third, coverage-corrected momentum-space evaluator independently reproduces the value to  $\sim 13$ – $14$  digits and then plateaus, confirming the anchor's leading digits; no  $(s, t)$ -frame quadrature reached further, and the definitive digits were ultimately supplied by the  $(k, w)$  refactorization (Eq. (??)), not by the modular route. *Second*, the guarded searches above used a *period-only* ring. A length-two, weight- $\leq 3$  iterated integral of weight-two Eisenstein series over  $X(2)$ —the shape forced by the two Feynman integrations—generically produces also the Legendre *quasiperiod*  $1/K(\frac{1}{2})$  (the second-kind companion,  $E(\frac{1}{2}) = \pi/4 K(\frac{1}{2}) + K(\frac{1}{2})/2$ ) and the weight-two Eisenstein  $L$ -value  $G = \beta(2) = L(2, \chi_{-4})$  (Catalan's constant; native to this family because the fibre period *is* the  $\mathbb{Q}(i)$  theta series:  $K(\lambda) = \frac{\pi}{2} \theta_3^2$  with  $\theta_3^2 = 1 + \sum_{n \geq 1} r_2(n) q^n$ ,  $r_2(n) = 4 \sum_{d|n} \chi_{-4}(d)$  by Jacobi's two-squares theorem—the weight-one, level-4 Eisenstein series of character  $\chi_{-4}$ , with  $L(\theta_3^2, s) = 4 \zeta(s) \beta(s)$ ). The often-quoted  $\int_0^1 K(k) dk = 2G$  holds in the *modulus* convention only; in this paper's parameter convention—which is also the physical  $d\rho$  measure of the co-area reduction below— $\int_0^1 K dm = 2$  and  $\int_0^1 E dm = \frac{4}{3}$ , both rational, so the convention-free reason  $G$  enters is the  $\chi_{-4}$  theta series, not those integrals [**MEASURED; test\_paper59\_theta\_chi4.py**]), neither of which lies in  $\{1, \pi, K(\frac{1}{2}), P_8\}$ . The conductor-4 arithmetic is thus carried by the fibre period at *every* fibre—a property of the family, not of the  $\tau = i$  point—and the physical Bessel twist sharpens it: the decay rate  $A = \sqrt{c_1} + \sqrt{c_2} = \sqrt{u}(1 + \sqrt{\rho})$  makes  $\sqrt{\rho} = \sqrt{1 - \lambda} = \theta_4^2/\theta_3^2$ —a  $\Gamma(4)$ -level modular function—physical, promoting the level from  $\Gamma(2)$  to  $\Gamma(4)$ , whose field is  $\mathbb{Q}(\mu_4) = \mathbb{Q}(i)$ , with  $4 = |\text{disc } \mathbb{Q}(i)| = \text{cond } \chi_{-4}$  [**MEASURED;***



`test_paper59_theta_chi4.py`]. In the corrected ring  $\{\pi, K(\frac{1}{2}), 1/K(\frac{1}{2}), G\}$  the weight- $\leq 2$  search is decisively negative, and the period-only weight- $\leq 3$  search is negative, so the finite closed form, should it exist, is *weight-three and requires  $G$* —and it is then the closed form of the  $D \rightarrow 0$  shadow, the residual question being whether the physical  $D = 1$  value itself collapses onto that ring element (since answered in the negative at clean heights by the 64-digit decoy-calibrated search above). This sharpened target also recasts the earlier merely-“consistent-with” negative as a ring-incompleteness artifact rather than, by itself, evidence of non-classicality: the non-classicality is carried separately, by the fibre’s irregular twist—established symbolically at the operator level in Sec. ?? (Newton polygon, Poincaré rank one) and witnessed numerically below—not by the integer-relation search.

[**OBSERVATION**] This places the three-center integral inside the elliptic (genus-one) layer of the cosmic-Galois / mixed-elliptic-motive structure that the Geometric Vacuum program’s Tannakian substrate reaches for [? ? ]—the concrete elliptic period the mixed-Tate substrate could not itself supply. [**OPEN**] The natural next step is therefore to place it in the Bessel-moment period algebra—to test whether it obeys a Broadhurst–Mellit quadratic relation or reduces to a critical  $L$ -value of the associated modular form [? ? ? ]—of which the explicit  $\Gamma(2)$  iterated-Eisenstein realization [? ? ] captures only the regular  $D \rightarrow 0$  shadow, both of a piece with the finite closed form of Sec. ???. We note the level is  $\Gamma(2)$ —the classical *universal Legendre* level, below the two-loop sunrise’s  $\Gamma_1(6)$ ; landing on the Legendre family is expected for a four-branch-point curve, so the content here is the rational modulus map and the in-domain complex-multiplication fibres, not an exotic level.

[**OBSERVATION**] The route to that closed form is now specific. The fibre period is the  $\Gamma(2)$  modular quantity  $K(m) = \frac{\pi}{2} \theta_3(0, q)^2$  at  $\tau = iK'(m)/K(m)$  (with  $\lambda(\tau) = m$ , both validated to 40 digits), so the modulus integration pulls back to the  $\tau$ -plane with the weight-two Jacobian in closed form,  $d\lambda/d\tau = i\pi \theta_2^4 \theta_4^4 / \theta_3^4 = i\pi \lambda(1-\lambda) \theta_3^4$  (the classical modular-lambda derivative; leading  $q$ -expansion  $i\pi(16q - 256q^2 + 2112q^3 - \dots)$ ) [**MEASURED**,  $\sim 10^{-32}$  **finite-difference** + **symbolic**; `test_paper59_bd_jacobian.py`]. The *physical* fibre, however, carries the  $D = 1$  exponential (Bessel) twist the pure period lacks—it degenerates to  $K_0(1)$ , not a modular period, and this irregularity is measured directly at the level of the master period itself: its large- $D$  asymptotic series is Gevrey-one (the optimal-truncation order grows linearly with  $D$  and the series then diverges) with Borel radius 2, matching the dominant singularity of the connection (??) [**MEASURED**; `test_paper59_resurgence_corner.py`, with the **irregular-connection witness** `test_L4_irregular_at_infinity`]*—so the integrated observable is a  $\Gamma(2)$  resurgent Lambert series rather*

than a bare modular period, the pure  $\Gamma(2)$  multiple modular value being its regular  $D \rightarrow 0$  shadow. This is exactly the class for which Broadhurst and Dorogoni give the iterated-Eisenstein / Lambert-series machinery for sunrise and banana integrals [? ]. With the Jacobian above in closed form the length-two-to-length-one reduction is now explicit: a co-area change of variables (modulus  $\rho = c_2/c_1$  outer, scale  $u = c_1$  inner) collapses the observable to a single  $X(2)$ -locus integral  $T2 = (16/\pi) \int_0^1 \Phi(\rho) d\rho$ , folded by the exact symmetry  $\Phi(\rho) = \Phi(1/\rho)/\rho^2$  [**MEASURED**, **reduction**  $2.2 \times 10^{-9}$  **vs the anchor**, **symmetry**  $5 \times 10^{-29}$ ; `test_paper59_coarea_reduction.py`]. The twist whose Eichler integral this is, is the *scale-integrated two-mass* fibre  $\Phi(\rho)$ —the one-mass  $N(D) = \int_1^\infty e^{-Dx}/\sqrt{Q} dx$  is only its  $\rho \rightarrow 0$  slice—and its cusp sits at the diagonal  $c_1 = c_2$ , where the curve degenerates to genus zero, so the leading Fourier–Whittaker (log) coefficient is the genus-zero diagonal value  $A = J(\frac{1}{4}, \frac{1}{4}, 1)/4$  (with the exact relation  $b_1 = -A$ ), one transcendence level below the observable. The cusp expansion  $\Phi = \sum_m [b_m \lambda^m \ln \lambda + e_m \lambda^m]$  is *asymptotic*: its leading reliable terms overshoot (68%  $\rightarrow$  51% of  $T2$ , non-monotone), which excludes any convergent low-order closure and leaves the *Borel–Lambert resummation* to the second cusp—the Stokes / median-summation data one level beyond the coefficients (the BD  $\sum_n a(n)q^n/(1-q^n)$  form)—as the single remaining specialist step.

[**SYMBOLIC** + **MEASURED**] The fold is itself a modular statement, and it locates the observable’s modular home one level lower than the family’s. With  $c = s(1-s)$  invariant under  $s \mapsto 1-s$ , the full symmetry group of the Feynman square acts on the modulus  $\rho = c_2/c_1$  through exactly  $\{\text{id}, \rho \mapsto 1/\rho\}$ ; the physical  $s \leftrightarrow t$  density exchange is  $\rho \mapsto 1/\rho$ , i.e.  $\lambda \mapsto \lambda/(\lambda-1)$ , the coset  $T\Gamma(2)$ , whose elliptic representative  $M = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$  satisfies  $M^2 = -I$  and  $M \equiv T \pmod{2}$ , and  $\langle \Gamma(2), M \rangle = \Gamma_0(2)$ . The folded co-area integral  $\int_0^1 \Phi d\rho$  is therefore an integral over a fundamental domain for the extended group: *the observable lives on  $X_0(2) = X(2)/\langle \iota \rangle$ , not on  $X(2)$*  (`tests/test_paper59_gamma0_2.py`). Two consequences, one positive and one deflationary. Positive:  $\Gamma_0(2)$  has a unique elliptic point, of order two with  $M^2 = -I$ —disc  $-4$ ,  $j = 1728$ , CM by  $\mathbb{Z}[i]$ —so the conductor-4 arithmetic attaches to the observable’s modular home canonically, complementing the fibre-independent  $\theta_3^2$  leg above. Deflationary: that elliptic point sits at  $\lambda = 2$  ( $\rho = -1$ ), the ramification point of  $X(2) \rightarrow X_0(2)$ , *off* the physical contour  $\rho > 0$ ; the involution’s physical fixed locus is the diagonal  $\rho = 1$ , the pure-Tate cusp; and  $\rho \mapsto 1-\rho$  is not induced by any physical symmetry, so the CM fibre  $\rho = \frac{1}{2}$  is one member of the  $s \leftrightarrow t$  orbit  $\{\frac{1}{2}, 2\}$ —a visited fibre, not a pinned one. (That a  $\mathbb{Z}_2$  symmetry of a modular family pins a  $\mathbb{Z}[i]$  point at its elliptic fixed locus is  $\text{PSL}_2(\mathbb{Z}) \cong \mathbb{Z}_2 * \mathbb{Z}_3$  torsion—order two forces disc  $-4$ , order three disc  $-3$ —so the arithmetic content of the involution is one bit, not a mechanism.)

[**MEASURED**] Two literature templates fix that step precisely and, between them, locate the one genuinely-open corner. A single-scale reduction of exactly this kind—a Bessel moment whose argument is a modular function, equated to a weight-two modular form and thence to an Eichler integral of a higher-weight form—is Zhou’s [?] (e.g.  $\int_0^\infty J_0(\alpha(z)t) I_0(t) K_0(t)^3 t dt = \frac{\pi^2}{16} Z_{6,3}(z)$  at level  $\Gamma_0(6)$ , and the weight-four Eichler representation of  $\int I_0 K_0^4$ ), whose special values close to complex-multiplication  $\Gamma$ -values or critical  $L$ -values. The resummation of the resulting cusp series across the two Fricke-dual cusps is Broadhurst–Dorigoni’s, both in the divisor-sum form they give for sunrise/banana integrals [?] and—closer to our setup—in the character-twisted precursor [?], whose transseries relates the value at one cusp to the same Lambert family at the Fricke-dual argument  $\tau \rightarrow -1/(r_1 r_2 \tau)$ , median-resummed with algebraic (character) Stokes data; the general engine—a Borel tower whose Stokes constants are  $L$ -function coefficients, resummed to a quantum modular form—is the modular-resurgence paradigm of Fantini–Rella [?] and the Eichler-integral median resummation of McSpirt–Rolen [?]. Our object sits one axis beyond each: the fibre (??) is a *two-scale* ( $a_1 \neq a_2$ ) Bessel moment where Zhou’s are single-scale, and its cusp coefficients are Bessel–Fourier–Whittaker coefficients rather than the Dirichlet divisor sums those resummations twist by. The distinction is concrete already at the leading coefficient: the diagonal value  $A = J(\frac{1}{4}, \frac{1}{4}, 1)/4 = 0.0790540321168\dots$  is itself a single-scale Bessel moment—by Eq. (??) and its  $t$ -derivative  $\int_0^\infty \cos(kb) e^{-t\sqrt{ck^2+1}} dk = 2t K_1(\sqrt{t^2+4b^2})/\sqrt{t^2+4b^2}$ ,  $A$  reduces (exactly) to a  $\sigma$ -integral of  $K_1$  at argument  $\sqrt{t^2+4\sigma^2}$ —and a guarded, decoy-controlled integer-relation search finds *no* low-height closure for it in the weight-one two-center ring  $\{\pi, e^{-a}, K_0, K_1, E_1\}$  at the natural arguments  $a \in \{2, 2\sqrt{2}\}$  [**MEASURED**; `test_routeC_momentum.py::test_paper59_diagonal_A`]. So even the leading cusp datum is an irreducible Bessel moment, not an elementary or divisor-sum quantity—consistent with a resurgent Lambert series whose individual coefficients need not be elementary for the resummed value to be a clean  $\Gamma(2)$  period. [**OPEN**] The exact remaining object is therefore the weight-two Eisenstein Eichler integral over  $X(2)$  against this two-scale Bessel twist—the two-scale generalization of the single-scale Bessel-moment theory of Fresán–Sabbah–Yu [?], itself open—covered by none of these templates individually but pinned by all of them jointly.

[**MEASURED**] The second-cusp resurgence itself is now characterized in closed form. The scale-integrated twist’s one-mass expansion ( $c_2 \rightarrow 0$ , one scale collapsing) is Gevrey  $(2n)!$ -type, and its dominant Borel singularity—the Stokes-data *location*—is  $z^* = -(\sqrt{c_1} \mp ib)^2 = (b^2 - c_1) \pm 2i\sqrt{c_1}b$ , with  $|z^*| = c_1 + b^2$  and trans-series action  $\sqrt{z^*} = b + i\sqrt{c_1}$  (the source sepa-

ration  $b$  and the scale  $\sqrt{c_1}$  fused into one complex action). It is a  $(z^* - w)^{3/2}$  branch (exponent  $p = -\frac{5}{2}$ , measured cleanly) with amplitude in *closed form*,  $2|A| = (c_1 + b^2)^{3/2}/(4\sqrt{\pi}\sqrt{c_1}b)$  (algebraic  $\times 1/\sqrt{\pi}$ , the  $1/\sqrt{\pi}$  from the  $(1 + \varepsilon)^{-5/2}$  boundary branch)—derived from the moment tail rate  $\sqrt{c_1} - ib$  and validated at three parameter points (fit-to-closed-form ratio 0.988–0.989, a uniform finite-order deficit converging to 1). This is the two-scale analog of the algebraic-Stokes closure of  $N(D)$  in Sec. ??, and the median-Borel resummation across the (complex, off-contour) singularity pair is unambiguous. [**OPEN**] The residual step to the finite value is the integration of this algebraic-Stokes trans-series over the Feynman family (the inner scale) and the modulus—the concrete form the two-scale Broadhurst–Dorigoni resummation takes here. Brute high-precision quadrature is not a route *in the  $(s, t)$  or co-area frames* (complex-singularity- and fibre-cost-limited there); the  $(k, w)$  refactorization Eq. (??) removes exactly that obstruction and supplies the *digits*, while the finite *closed form* remains this resummation question.

[**MEASURED**] *The Stokes location is a complexified light cone.* In position space each packet is analytic in  $b^2$  with branch points at the endpoint of its source-time superposition, where the Euclidean interval closes,  $b^2 + p_i^2 = 0$ ; the  $j_0$  smearing moves the singularity to the endpoint pinch  $\beta^* = \pm|\mathbf{W}| \pm ip_1$ , so the Borel location above is the squared complexified interval,  $z^* = (\beta^*)^2$ ,  $|z^*| = p_1^2 + |\mathbf{W}|^2$ , of which  $-(\sqrt{c_1} \mp ib)^2$  is the  $D = 1$  specialisation  $p_1 = \sqrt{c_1}$ ; the trans-series action  $a_2\sqrt{z^*}$  is the heavy mass times that interval. Moment growth of the position-space fibre confirms modulus and phase at eight  $(c, |\mathbf{W}|, D)$  points, including  $D \neq 1$ , where the interval form and the  $D$ -independent form differ by 35% [**MEASURED**, 0.6–5% from twelve coefficients]. Two consequences. The median resummation is unambiguous because a Euclidean interval never vanishes at real separation: the absence of a Stokes ambiguity is Euclidean positivity. And the  $(s, t) \rightarrow (0, 0)$  corner is the coincidence (ultraviolet) point, where the interval closes and the Borel singularities collide at the origin—which is why that corner, and only that corner, obstructs the family integration. There the masses diverge at fixed ratio, so the modulus survives and the Duffy angle *is* the modulus; the leading behaviour is the ultraviolet power count  $J = \sigma^3 A(a) + O(\sigma^5)$ ,  $\sigma^3 = m_1^{-2} m_2^{-2} m$ , so the  $\rho^{3/2}$  non-analyticity is an odd mass dimension and carries no  $\pi$ . At the three corners with  $|\mathbf{W}| \neq 0$  the oscillation survives, a Dirichlet limit supplies one further power and an explicit  $\pi$ , and the integer-leading behaviour is now closed:  $J = \sigma^4 \frac{\pi}{2} G(1)^2 \alpha(1 - \alpha)/|\mathbf{W}| + O(\sigma^5)$  with  $G(1) = 7/e$  [**MEASURED**,  $10^{-4}$  with  $O(\sigma^2)$  approach, four corner/angle combinations; `test_paper59_euclidean_dictionary.py`].

[**OBSERVATION**] Read against the two-layer taxonomy, the object separates on two kinematic axes: the mass split turns on the elliptic (genus-one) level and

the Euclidean-time displacement turns on the irregular level—the curve carries no  $D$ , and the large- $D$  series is Gevrey-one at every  $\rho$ , including the genus-zero slice  $\rho = 0$  where the fibre is  $K_0(D)$ —while the modulus then also grades the irregular data through the Borel radius  $\min(2, a_2/a_1)$ , the smallest difference of the four exponential rates. Every  $\pi$  in the object comes from the momentum-space measure or from integration over the momentum half-line; the non-compact Euclidean-time direction contributes no  $\pi$  but the exponential and its Stokes data, consistent with the compactification reading of Paper 35 [? ]. (In Euclidean signature the labels are structural, not metric:  $D$  is the shift each propagator carries,  $b$  the shared coordinate that is integrated out; the temporal reading is offered as a reading.)

**[OBSERVATION]** The algebraic-Stokes closures of this paper sit inside a pattern that now spans the corpus’s two-electron integral classes. Four objects have had their resurgent data computed: the Paper-18 exchange-class seed  $e^a E_1(a)$  (Borel transform  $1/(1+\zeta)$ , one simple pole, Stokes constant  $2\pi i \cdot 1$ ); the Paper-58 hybrid two-center class  $\{E_1, \ln\}$  (all Borel singularities at  $\pm a_d$ , the far-center exponent; Stokes constants  $2\pi i \times \mathbb{Q}(\sqrt{d})$ , extracted blind from large-order coefficients to 35–81 digits and matched by exact integer relations; and the sector constants cancel pairwise, so the physical integral is *cut-free* in  $R$ —the median resummation *is* the closed form, `tests/test_paper59_resurgent_skeleton.py`); the one-mass fibre  $N(D)$  of Sec. ?? (algebraic over  $\mathbb{Q}(\rho)$ ); and the second-cusp triple above (algebraic  $\times 1/\sqrt{\pi}$ ). In all four the connection data is algebraic over the parameter field up to a  $\pi$ -power that is itself fixed by the Borel singularity type (simple pole  $\rightarrow 2\pi i$ ; square-root branch  $\rightarrow \pi^0$ ; three-halves branch  $\rightarrow 1/\sqrt{\pi}$ )—nothing in the Stokes data is free. What varies is the *boundary period*: absent for the seed;  $\ln \Lambda$  with  $\Lambda \in \mathbb{Q}$  the multiplicative cross-ratio of the four Borel actions—i.e. *forced by the skeleton*—for the hybrid class; the elliptic  $K(1-\rho)$ , *not* determined by the Stokes data, for  $N(D)$ ; and the open  $\Gamma(2)$  object for  $T2$ . The forced  $\rightarrow$  free transition of the boundary period thus sits between weight one and genus one—the same seam at which the closed-form program of this paper opened. The named next test—the exchange class, whose  $R$ -dependent logarithm makes its Borel singularity logarithmic rather than polar—has since been run and the pattern stands at **five for five**: the exchange kernel collapses to a single-sector reduced normal form whose Borel transform is the closed-form, charge-neutral sum  $\mathfrak{B}(\xi) = \frac{1}{2ab} \sum_j q_j (\xi - \xi_j) \ln(\xi - \xi_j)$  over the four rational positions, with *integer* Stokes charges  $q_j \in \{-1, +1, +1, -1\}$  (no radical needed, unlike the hybrid’s  $\sqrt{d}$ ); the boundary log is forced even more tightly than the hybrid’s cross-ratio—its argument is  $\kappa = \prod_j \lambda_j^{q_j} = 2ab/(a+b)$ , the exponents *being* the Stokes charges; and Euler’s  $\gamma$  never exists in the Borel plane at all: the  $R$ -dependent log moves one Borel singularity onto the sector’s own origin (pole  $\rightarrow u \ln u$  branch), and  $\gamma$  is exactly the bookkeeping cost of writing that origin singularity in

the  $R$  variable—which finally tags the  $\gamma$  of the weight-one seed set as coordinate bookkeeping, not a projection transcendental. The object is Borel-summable with no lateral ambiguity yet genuinely multivalued in complex  $R$  (the price of the  $R$ -dependent log, costing no new transcendental), where the hybrid class was single-valued (`tests/test_paper59_resurgent_skeleton.py`). The  $\pi$ -power law refines accordingly: an *integral* local Borel exponent (pole *or* log branch) fixes  $2\pi i$ ; a half-integral one fixes  $\pi^0$  or  $1/\sqrt{\pi}$ . (Read against the two-layer taxonomy this is a candidate refinement—a resurgent-skeleton grading *within* the transcendental layer—recorded as an observation, now on five objects; the residual open leg is the heteronuclear infinite  $\tau$ -sum.)

## VIII. THE BESSEL-MOMENT PERIOD ALGEBRA: MASTER FAMILY AND QUADRATIC RELATIONS

Section ?? places the integrated observable in the Bessel-moment class; we now make that placement quantitative by exhibiting the period algebra it inhabits. The rank-four irregular connection (??) has four master periods, the Laplace transforms

$$s_c(D) = \int_{\gamma_c} \frac{e^{-Dx} dx}{\sqrt{Q(x)}}, \quad Q(x) = (x^2 - 1)(\rho x^2 + 1 - \rho), \quad (16)$$

over the four independent cycles  $\gamma_c$  joining the branch points  $\{\pm 1, \pm i\sqrt{(1-\rho)/\rho}\}$ : the physical moment  $s_K$  (the cut  $[1, \infty)$ , which is  $N(D)$  itself,  $\sim e^{-D}$ ), its companion  $s_I$  ( $[-1, 1]$ ,  $\sim e^{+D}$ ), the imaginary-axis period  $s_J$  ( $\sim \cos \omega D$ ), and the sector  $s_Y$  carrying the  $D \ln D$  non-analyticity. **[MEASURED]** The three regular masters satisfy Eq. (??) to  $\sim 10^{-15}$  across  $\rho$  and  $D$  (each annihilated:  $s_K \sim 10^{-18}$ ,  $s_I \sim 10^{-17}$ ,  $s_J \sim 10^{-15}$ ).

*The Wronskian.* **[SYMBOLIC]** The operator (??) is formally self-adjoint (Sec. ??), so its differential Galois group lies in  $\mathrm{Sp}_4$ , and Abel’s identity—with the subleading coefficient ratio  $p_3/p_4 = 2/D$ —fixes the Wronskian of the master matrix exactly as a pure monomial,

$$W(D) = W_0 D^{-2} : \quad (17)$$

the transcendence of the individual masters cancels in their determinant, the elliptic analogue of the classical  $W[K_0, I_0](D) = 1/D$ .

*Quadratic period relations.* **[MEASURED, 25 digits]** Self-adjointness makes the Lagrange bilinear concomitant  $B[s_a, s_b]$  constant in  $D$ ; evaluated on the masters it is moreover independent of the modulus  $\rho$  and lands on rational multiples of  $\pi$ ,

$$B[s_K, s_I] = -\pi, \quad B[s_K, s_J] = 0, \quad B[s_I, s_J] = 2\pi, \quad (18)$$

verified to 25 digits at  $D = 1$  across the tested  $\rho$ , and  $D$ -independent by the self-adjoint Lagrange identity. These are the Broadhurst–Mellit / Fresán–Sabbah–Yu quadratic relations between Bessel moments [? ? ?

], here for the  $\Gamma(2)$  (Legendre) family—the elliptic lift of  $W[K_0, I_0] = 1/D$ . Structurally the concomitant is the de Rham–Betti intersection pairing of the thimble cycles times the branch-point half-residue  $\pi$  (the square-root  $(-1)$ -monodromy of  $1/\sqrt{Q}$  at each simple branch point, the elliptic lift of the  $1/\pi$  normalizing  $I_0$  and of  $W[K_0, I_0] = 1/D$ ), so  $B = \pi J$  with  $J$  an integer intersection form.

**[SYMBOLIC]** *The intersection form, identified.* We close the identification of  $J$  from the branch-point asymptotics—no numerics enter in this leg. Near a simple branch point  $Q \sim Q'(x_c)(x - x_c)$ , so each master has the leading Laplace behaviour  $s_c(D) \sim e^{-x_c D} \sqrt{\pi/(Q'(x_c)D)}$  (the half-order integral  $\int_0 e^{-Dt} t^{-1/2} dt = \Gamma(\frac{1}{2})/\sqrt{D}$ ); since the concomitant is exactly constant in  $D$ , its value equals its  $D \rightarrow \infty$  limit, which these leading terms capture (the subleading  $O(1/D)$  corrections vanish in the limit). In the Lefschetz-thimble basis  $\{\gamma_{+1}, \gamma_{-1}, \gamma_{+i\omega}, \gamma_{-i\omega}\}$ —one thimble per branch point, the basis in which the monodromy  $M_0$  of Sec. ?? is the exact integer matrix—two facts follow. *Block-diagonality:*  $s_a s_b \sim e^{-(x_a + x_b)D}$ , so a nonzero *constant* forces  $x_a + x_b = 0$ ; the branch points pair as  $\{+1, -1\}$  (the real  $K_0/I_0$  sector) and  $\{+i\omega, -i\omega\}$  (the imaginary  $J_0/Y_0$  sector), and every cross-sector pairing ( $x_a + x_b \neq 0$ ) vanishes identically. *Plane-equality and the unit  $\pi$ :* on a within-sector pair the leading concomitant is  $B[s_x, s_{-x}] = 2x(-2\rho x^2 + 2\rho - 1)A_x A_{-x}$  with  $A_x A_{-x} = \pi/\sqrt{Q'(x)Q'(-x)}$ , and the exact identity  $Q'(x)Q'(-x) = -4x^2(2\rho x^2 - 2\rho + 1)^2$  cancels the algebraic factor completely, leaving

$$B[s_x, s_{-x}]^2 = -\pi^2 \quad \text{independently of } x \text{ and } \rho, \quad (19)$$

so *both* symplectic planes carry the same pairing, of magnitude exactly  $\pi$  (the  $\pi = \Gamma(\frac{1}{2})^2$  of the two branch-point half-integrals). Hence

$$J = \Omega := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad B = \pi \Omega, \quad (20)$$

the canonical block symplectic form; in the period-cut basis this is the real  $-\pi, 0, 2\pi$  measured above (a numerical thimble normalization instead reads  $B = \pi(\rho i)\Omega$ , the same  $\Omega$ ). This  $\Omega$  is *forced*, not fitted: the antisymmetric forms preserved by the monodromy,  $M_0^T J M_0 = J$ , constitute a four-parameter family, and imposing the concomitant’s vanishing cross-sector pairings selects *exactly*  $\mathbb{Z}\Omega$ . It is unimodular ( $\det \Omega = 1$ , hence non-degenerate), and  $M_0^T \Omega M_0 = \Omega$  holds identically over  $\mathbb{Z}$ , so the differential Galois group of Eq. (??) lies in the *integral* symplectic group  $\mathrm{Sp}(\Omega, \mathbb{Z}) = \mathrm{Sp}_4(\mathbb{Z})$ —the concrete, integral sharpening of the self-adjointness that placed it in  $\mathrm{Sp}_4$ . *Tier scope.* The asymptotic identification above is numerics-free, but every clause invoking  $M_0$ —the forcing argument and the  $\mathrm{Sp}_4(\mathbb{Z})$  containment—inherits that matrix’s **[SYMBOLIC + MEASURED]** status:  $M_0$  is an exact integer certificate rather than a floating-point fit, but it was

obtained by computation, not derived in closed form. The three physical period cuts  $\{[1, \infty), [-1, 1], i[-\omega, \omega]\}$  realize only a rank-two sub-block of  $\pi\Omega$  (the measured  $-\pi, 0, 2\pi$ ); it is the fourth,  $Y_0$ -sector master—the one carrying the  $D \ln D$ —that completes the pairing to the full nondegenerate rank-four form. The symbolic identification of the intersection form is thus complete:  $B = \pi\Omega$ , the Broadhurst–Mellit / Fresán–Sabbah–Yu quadratic relations realized for the  $\Gamma(2)$  family, backed by `tests/test_routeCmomentum.py` (`test_intersection_form_pi_and_planes_are_symbolic` for Eq. (??) and the block structure, `test_intersection_form_is_forced_canonical_symplectic` for the  $M_0$ -forcing, `test_intersection_form_period_cut_values` for the  $-\pi, 0, 2\pi$  values, and `test_paper59_bessel_moment_algebra.py::test_paper59_period` witnessing them to the quoted 25 digits).

*Not a cusp-form  $L$ -value.* **[OBSERVATION]** Because  $M_\bullet(\Gamma(2)) = \mathbb{C}[\theta_2^4, \theta_4^4]$  is free on two weight-two generators,  $\dim S_k(\Gamma(2))$  vanishes for  $k = 2, 4$  and first reaches 1 at  $k = 6$  (the cusp form  $\theta_2^4 \theta_3^4 \theta_4^4$ ). The observable’s transcendental weight is at most three (length  $\leq 2$  over  $X(2)$ , from the two Feynman integrations), whereas a weight-six cusp form’s critical  $L$ -values sit at motivic weight five; and the period pairing just exhibited is Eisenstein-flavoured (pure  $\pi$ -rationals), not cuspidal. The integrated three-center integral is therefore *not* a critical  $L$ -value of a  $\Gamma(2)$  cusp form: it is an Eisenstein / complex-multiplication period, built from  $\pi$  and the CM-fibre  $\Gamma$ -values of Sec. ?? ( $K(\frac{1}{2}) = \Gamma(\frac{1}{4})^2/4\sqrt{\pi}$ , the discriminant-8 period, and their kin). **[MEASURED]** A guarded integer-relation search (with a same-magnitude decoy control) confirms the value is not a low-height combination of the small weight- $\leq 2$  disc-4 Eisenstein/CM ring, nor of the small weight- $\leq 1$  disc-8 ring  $\{1, \pi, \varpi, P_8\}$  ( $P_8$  the discriminant-8 period)—both *decisive* at the conservatively cross-validated  $\sim 16$  digits: two structurally independent evaluators (the spectral corner of Sec. ?? and a decay-scaled-fibre quadrature) agree to 16 digits, the corner evaluator extending single-method to  $\sim 19$ ; on the decisive rings the relation height is far above low and tracks the decoy at every tested precision. The wider searches—through weight three in disc-4, and the disc-8 ring at weight two—remained only *consistent* with the negative: those dimension-ten bases are over-determined at this precision (target and decoy both turn up small but precision-*unstable* relations, changing from one precision to the next). The decision has since been made at 64 digits via the  $(k, w)$  refactorization (Eq. (??)): *decisively negative* through weight three, including the Catalan-completed corrected ring, at height  $\leq 10$ —so the finite closed form, if it exists at all, either has height  $> 10$  in that ring or lives outside it.

This narrows the open problem sharply: the finite closed form, should it exist, is a reduction of the observable to  $\pi$ , the CM-fibre  $\Gamma$ -values, their Legendre *quasiperiods*, and the weight-two Eisenstein  $L$ -value

$G = \beta(2)$  (Sec. ??), within the Bessel-moment period algebra—a two-scale, shifted member of the Broadhurst–BBBG class [? ? ] evaluated over  $X(2)$ —and is neither an elliptic dilogarithm nor a cusp-form  $L$ -value. **[OBSERVATION]** Because the corrected-ring searches are now decisively negative through weight three at clean heights (64 digits, height  $\leq 10$ ), any such finite combination either belongs to the  $D \rightarrow 0$  *shadow*—where the shadow reading of Sec. 4 places the weight-three  $G$ -requiring element—or sits at height  $> 10$  in that ring; the physical value itself carries no measured conductor-4 content at reachable heights.

## IX. EXPLICITLY-CORRELATED INTEGRALS ARE NATIVE TO THIS REPRESENTATION

The momentum reduction of Sec. ?? is *kernel-agnostic*. A two-electron matrix element of *any* operator  $f(r_{12})$  between orbital products is, by the convolution theorem,

$$\langle \phi_a \phi_b | f(r_{12}) | \phi_c \phi_d \rangle = \frac{1}{(2\pi)^3} \int d^3k \tilde{\rho}_{ac}(\mathbf{k})^* \tilde{f}(\mathbf{k}) \tilde{\rho}_{bd}(\mathbf{k}), \quad (21)$$

the identical object as the electron-repulsion integral with the Coulomb kernel  $(1/r_{12}) = 4\pi/k^2$  replaced by  $\tilde{f}$ , and the angular closure to the spherical Bessel kernel of Eq. (??) applies unchanged. Explicitly-correlated (F12) methods take  $f$  to be a correlation factor; the standard choice is the Slater geminal  $f = e^{-\gamma r_{12}}$ , whose transform  $\tilde{f}(k) = 8\pi\gamma/(k^2 + \gamma^2)^2$  is exact, elementary, and—unlike the Coulomb kernel—*regular* at  $k = 0$ . In this representation the geminal is therefore a native object: no expansion of  $e^{-\gamma r_{12}}$  in Gaussians (the first workaround of Gaussian-basis F12) is required. **[MEASURED]** The momentum form (??) reproduces a direct position-space quadrature to  $10^{-14}$ , and the Coulomb self-repulsion of a  $1s$  density to its exact value  $5a/8$  (`test_f12_kernel_swap_and_native_geminal`).

Whether the integral closes in elementary functions is governed *not* by the operator but by the *density topology*. **[SYMBOLIC]** When both transition densities are single-center the radial integrand is rational and the pair integral is genus zero: e.g.  $\int_0^\infty [k^2/(k^2 + \gamma^2)^2] j_0(kR) dk = (\pi/4\gamma) e^{-\gamma R}$  (`test_f12_geminal_pair_is_elementary`). A two-center transition density, however, contributes a Fock dispersion factor  $\sqrt{ck^2 + 1}$  (Sec. ??), and two of them build the very curve of Eq. (??),  $y^2 = (c_1 k^2 + 1)(c_2 k^2 + 1)$ , whose period is a complete elliptic integral (`test_f12_multicenter_shares_eri_elliptic_genus`). A rational kernel—Coulomb  $4\pi/k^2$  or geminal  $8\pi\gamma/(k^2 + \gamma^2)^2$  alike—is a rational function on this curve, so  $\tilde{f} dk/y$  is again an elliptic integral: the genus is *unchanged*, the geminal merely trading an elliptic integral of the first kind (the Coulomb period) for one of the third kind (its poles at  $k^2 = -\gamma^2$ ). The transcendence class is thus set by the number of two-center transition

densities—zero or one gives genus zero, two gives genus one—exactly as for the repulsion integral.

**[OBSERVATION]** The consequence is a *unification*, not an escape. Explicitly-correlated integrals are native to the Fock momentum representation and cost the same as the corresponding repulsion integrals, because here the difficulty of a two-electron integral lives entirely in the densities, never in the operator between them. In particular the elliptic obstruction of this paper—the genuine three-center, two-two-center-density integral—gates explicitly-correlated *evaluation* as well: the corresponding closed form (the kernel changes the kind of period, not the curve) that would evaluate the three-center repulsion integral would evaluate the multi-center F12 correction. What correlation buys is not a cheaper integral but a far more compact basis—a single  $r_{12}$  factor recovers the electron–electron cusp that orbital products approach only as  $\ell \rightarrow \infty$ —so the route trades many repulsion integrals for few correlated ones drawn from the same transcendence classes. We do not here treat the many-electron (resolution-of-identity) machinery of a full F12 method, nor a resource comparison with tuned Gaussian F12; the statement is about the two-electron integrals, whose momentum-space transcendence we have mapped.

## X. SCOPE, AND THE OPEN FRONTIER

*Scope.* The demonstration is for  $1s$  orbitals at a common exponent. Higher  $(n, \ell, m)$  multiply the integrand of Eq. (??) by solid-harmonic polynomials in  $\mathbf{k}$  and shift the exponents; these change the weights and arguments but not the genus of the curve (??), which is set by the two distinct scales. The result is a statement about the transcendence *class* of the three-center integral, established on its simplest genuine instance. It does not by itself deliver a polyatomic energy: a molecular Hamiltonian requires the full three-center tensor, and a value that is part closed-form and part numerical is neither more accurate (the Gaussian fits are already exact to fit quality) nor decidable. What the reduction buys is an accurate, fast, geometry-native evaluator for the three-center integral and—the point of this paper—a precise diagnosis of why it is hard.

*The open problem.* As shown in Sec. ??, the elliptic-dilogarithm route is obstructed:  $N(D)$  is not an elliptic dilogarithm of elementary or period-level data but a period of the rank-four irregular connection (??). We settle the differential-Galois *reducibility*: the fourth-order operator is irreducible over  $\mathbb{C}(D)$ —hence over  $\mathbb{Q}(\rho)(D)$ —because it is the Fourier–Laplace transform of an irreducible connection, confirmed concretely by its exact-integer monodromy (Sec. ??), so  $N(D)$  admits no closed form through any factorization of the operator. The one remaining question is whether the irreducible connection nonetheless carries a *transcendental* closed form, which we do not settle. Section ?? localizes it: the object is an Eisenstein / complex-multiplication Bessel-moment

period—not a cusp-form  $L$ -value and not an elliptic dilogarithm—whose master family obeys the quadratic period relations  $B = \pi \Omega$ ,  $\Omega$  the canonical symplectic intersection form now identified in closed form (Sec. ??), preserved by the integral monodromy; the open piece is the explicit finite reduction of the observable itself to  $\pi$  and the CM-fibre  $\Gamma$ -values. (This tempers a literature-scout expectation that the object would sit at the tractable elliptic-dilogarithm level because it has only two propagator factors; that pattern-match from the sunrise is overridden by the in-module computation of Sec. ??.) Because it continues the Averys’ momentum-space Sturmian program and requires elliptic-Feynman-integral technique one level beyond the sunrise, it is the natural place for collaboration.

## XI. CONCLUSION

The three-center electron-repulsion integral is not merely a harder version of the two-center one; it is an

object of higher genus. Two centers close at genus zero over  $\{E_1, \ln, \gamma\}$ ; three centers, read in momentum space where the coordinate obstruction dissolves, reduce to a two-scale Bessel moment on an elliptic curve, an elliptic deformation of the Bessel  $K_0$  with an explicit fourth-order Picard–Fuchs equation and a period that is a complete elliptic integral. The seventy-year difficulty of polyatomic Slater integrals is, at the level of transcendence, a genus jump; and a classical molecular integral turns out to be an instance of the modern theory of elliptic Feynman integrals—a bridge that, to our knowledge, had not been drawn. The finite closed form remains open—and, we have shown, past the reach of the sunrise’s elliptic-dilogarithm mechanism, whose inhomogeneity is absent here: the object is a period of a rank-four irregular connection which we prove irreducible—it is the Fourier–Laplace transform of an irreducible connection, and its monodromy admits no invariant coordinate subspace—so it does not close by any factorization. Whether it closes at the irreducible level—as an Eisenstein/CM Bessel-moment period, a finite reduction to  $\pi$  and the CM-fibre  $\Gamma$ -values (Sec. ??)—is the one frontier that remains, posed here as it deserves to be.

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